Step 5 : Calculation of Test Statistic

The value of Z for the given sample information is calculated from

$$z_0 = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \text{ as}$$

$$z_0 = \frac{30 - 27}{3/\sqrt{100}}$$

Thus, $z_0 = 10$.

Step 6 : Critical Value

Since H_1 is a one-sided (right) alternative hypothesis, the critical value at $\alpha = 0.05$ is $z_e = z_{0.05} = 1.645$.

Step 7 : Decision

Since H_1 is a one-sided (right) alternative, elements of the critical region are defined by the rejection rule $z_0 > z_e = z_{0.05}$. Thus, it is a right-tailed test. Since, for the given sample information, $z_0 = 10 > z_e = 1.645$, H_0 is rejected.

1.11 TEST OF HYPOTHESES FOR EQUALITY OF MEANS OF TWO POPULATIONS *(Population variances are known)*

Procedure:

Step-1 : Let μ_X and σ_X^2 be respectively the mean and the variance of Population -1. Also, let μ_Y and σ_Y^2 be respectively the mean and the variance of Population -2 under study. Here σ_X^2 and σ_Y^2 are known admissible values.

Frame the null hypothesis as $H_0: \mu_X = \mu_Y$ and choose the suitable alternative hypothesis from

(i) $H_1: \mu_X \neq \mu_Y$ (ii) $H_1: \mu_X > \mu_Y$ (iii) $H_1: \mu_X < \mu_Y$

Step 2 : Let $(X_1, X_2, \ldots, X_m)$ be a random sample of m observations drawn from Population-1 and $(Y_1, Y_2, \ldots, Y_n)$ be a random sample of n observations drawn from Population-2, where m and n are large(*i.e.*, $m \geq 30$ and $n \geq 30$). Further, these two samples are assumed to be independent.

Step 3 : Specify the level of significance, α.

Step 4 : Consider the test statistic $Z = \dfrac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{\sqrt{\dfrac{\sigma_X^2}{m} + \dfrac{\sigma_Y^2}{n}}}$ under H_0, where $\bar{X}$ and $\bar{Y}$ are respectively the means of the two samples described in Step-2.

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